

Assignment 3

Step 1: Problem Statement

Create a polymorphic array to store and work with animal objects.

For each line in the test file, Animals.txt, read the information about the animal.

Create a specific animal object based on the species, and place the animal object into the array. After the animal objects are

in the Array, perform the following 3 tasks on the array of animals

1. Call displayAnimal method to display all animals in the array

2. Call findClimbers method to find and display all animals that can climb

3. Call findMostSkilled method to find and display the most skilled animal (one with the highest combined speed in all skills).

~~Step 2: Problem Statement~~
~~Most Skilled~~

Step 2° What I know / Don't know

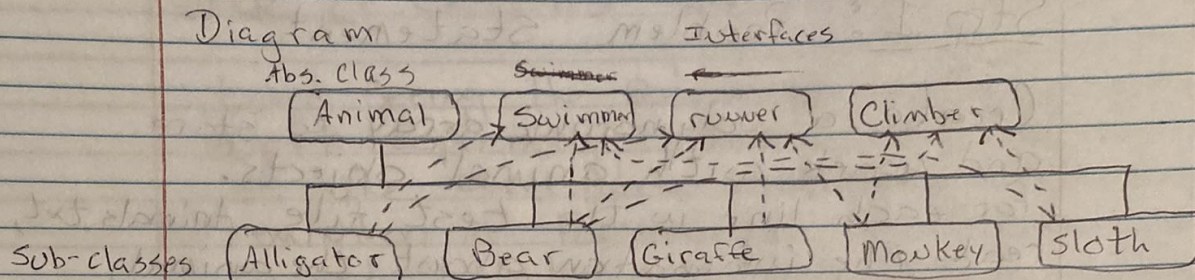
What I know

- Good with arrays
- Good with objects
- Good with arraylists
- Good with interfaces
- displaying info
- calling methods

What I don't know

- Casting

Step 3: ~~Write~~ Pseudocode main & find Most Skilled



Main

Read ray size from file int theSize = nextInt
Creat Animal[] animals = new Animal[theSize]

loop through file to create new
animal objects & place in Array

For $\leq$

readFile.next() = name

readFile.next() = Type

readFile.nextInt() = Swimspeed

nextInt() = run speed

nextInt() = climb speed

if $\emptyset$, don't send to constructor

Animal[i] = new Type(Type);

}

Call displayAnimal method();

Call findClimbers method();

Call findMostSkilled method(); returns
int Index

display most skilled from Array based on Index

Step 3 (cont.)

Find Most Skilled

create $\text{int}^+ \text{mostSkilledIndex} = 0;$

Loop through Array to establish most skilled

For $i = 1; i < \text{ArrayLength}; i++;$

if $\text{animal}[i].\text{swim} + i.\text{run} + i.\text{climb}$

$> \text{animal}[\text{mostSkilledIndex}].\text{swim} +$

$\text{run} + \text{climb}$

Then ~~mostSkilledIndex~~ $\{$

$\text{mostSkilledIndex} = [i]$

$\}$

return $\text{mostSkilledIndex};$

Step 4: What I Learned

Learned a lot about how to use interfaces and casting. Was a little challenging trying to get the casting down at first but once I had the right syntax it actually became pretty intuitive. Additionally, good exercise in keeping everything organized and how to layout the various aspects of the code in conjunction / relation to each other. It's a good example of why folks use other files for classes to keep things concise.